Homework 3

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CS 427

1) You have seen a 2-out-of-2 secret sharing scheme based on XOR. Generalize that scheme to a 3-out-of-3 scheme and prove that it is secure.

Starting with

Share(m)

S1←{0,1}^λ

S2←{0,1}^λ

$S3 = s1 \oplus s2 \oplus m$

Return (s1,s2,s3)

To prove this is secure I will show that $\mathcal{L}^{\Sigma}_{\text{tsss-L}} \equiv \mathcal{L}^{\Sigma}_{\text{tsss-R}}$ using a hybrid proof.

Starting with the left side and trying to prove that it is interchangeable with the right

Share(m_L,m_R,U):

If |U| >= 3 return err

S1←{0,1}^λ

S2←{0,1}^λ

$S3 = s1 \oplus s2 \oplus m_L$

Return (s1,s2,s3)

Next I will break the main body into sections of an if statement because the scheme generates s1/s2 differently than s3, separating like this will not affect the output

If |U| >= 3 return err

If(U={1})

    S1←{0,1}^λ

    S2←{0,1}^λ

    $S3 = s1 \oplus s2 \oplus m_L$

    Return(s1)

If(U={2})

    S1←{0,1}^λ

    S2←{0,1}^λ

$S3 = s1 \oplus s2 \oplus m_L$

Return(s2)

If(U={3})

S1 ← $\{0,1\}^\lambda$

S2 ← $\{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_L$

Return(s3)

If(U={1,2})

S1 ← $\{0,1\}^\lambda$

S2 ← $\{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_L$

Return(s1,s2)

If(U={1,3})

S1 ← $\{0,1\}^\lambda$

S2 ← $\{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_L$

Return(s1,s3)

If(U={2,3})

S1 ← $\{0,1\}^\lambda$

S2 ← $\{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_L$

Return(s2,s3)

Next, I change m_L to m_R in all the if blocks where u=3 is not included, this is because the way s1 and s2 are generated it does not depend on m at all, so we can change it to m_R

If |U| >= 3 return err

If(U={1})

S1 ← $\{0,1\}^\lambda$

S2 ← $\{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_R$

Return(s1)

If(U={2})

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_R$

Return(s2)

If(U={3})

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_L$

Return(s3)

If(U={1,2})

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_R$

Return(s1,s2)

If(U={1,3})

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_L$

Return(s1,s3)

If(U={2,3})

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_L$

Return(s2,s3)

Next in each if block that considers U=3 being included I decompose the $S3 = s1 \oplus s2 \oplus m_L$ to two lines. At the same time, because s1 and s2 are clearly still distributed randomly that means the first line is effectively

$S3 = s1 \oplus s2$

$S3 = S3 \oplus m_L$

At the same time, because s1 and s2 are clearly still distributed randomly that means $S3 = s1 \oplus s2$

Effectively the same as $k \leftarrow \{0,1\}^\lambda$

Resulting in

If $|U| >= 3$ return err

If($U=\{1\}$)

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_R$

Return(s1)

If($U=\{2\}$)

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_R$

Return(s2)

If($U=\{3\}$)

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$k \leftarrow \{0,1\}^\lambda$

$S3 = k \oplus m_L$

Return(s3)

If($U=\{1,2\}$)

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_R$

Return(s1,s2)

If($U=\{1,3\}$)

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$k \leftarrow \{0,1\}^\lambda$

S3 = $k \oplus m_L$

Return(s1,s3)

If(U={2,3})

S1 $\leftarrow \{0,1\}^\lambda$

S2 $\leftarrow \{0,1\}^\lambda$

k $\leftarrow \{0,1\}^\lambda$

S3 = $k \oplus m_L$

Return(s2,s3)

Now, each if block containing u=3 has been refactored to match the format of our OTP encryption. This means it can be replaced with eavesdrop(m_L,m_R) and then linked with the OTP OTS-L library. Using the composition lemma OTP OTS-L can be replaced with the R version because we have already shown that to be interchangeable. This results is

If |U| >= 3 return err

If(U={1})

S1 $\leftarrow \{0,1\}^\lambda$

S2 $\leftarrow \{0,1\}^\lambda$

S3 = $s1 \oplus s2 \oplus m_R$

Return(s1)

If(U={2})

S1 $\leftarrow \{0,1\}^\lambda$

S2 $\leftarrow \{0,1\}^\lambda$

S3 = $s1 \oplus s2 \oplus m_R$

Return(s2)

If(U={3})

S3 $\leftarrow$ eavesdrop(m_L,m_R) $\leftrightarrow$

$\mathcal{L}^{\text{OTP}}_{\text{ots-R}}$

EAVESDROP(m_L, m_R):

$k \leftarrow \{0,1\}^\ell$

$c := k \oplus m_R$

return c

Return(s3)

If(U={1,2})

S1 $\leftarrow \{0,1\}^\lambda$

S2 $\leftarrow \{0,1\}^\lambda$

S3 = $s1 \oplus s2 \oplus m_R$

Return(s1,s2)

If(U={1,3})

S1←{0,1}^λ

S2←{0,1}^λ

S3 ← eavesdrop(m_L,m_R) ↔

Return(s1,s3)

$\mathcal{L}^{\text{OTP}}_{\text{ots-R}}$

EAVESDROP(m_L, m_R):

$k \leftarrow \{0,1\}^\ell$

$c := k \oplus m_R$

return c

If(U={2,3})

S1←{0,1}^λ

S2←{0,1}^λ

S3 ← eavesdrop(m_L,m_R) ↔

Return(s2,s3)

$\mathcal{L}^{\text{OTP}}_{\text{ots-R}}$

EAVESDROP(m_L, m_R):

$k \leftarrow \{0,1\}^\ell$

$c := k \oplus m_R$

return c

Now bringing that library functionality back to inline we get all the m_L replaced with m_R and the left hand library has been shown to be interchangeable with the right

If |U| >= 3 return err

If(U={1})

S1←{0,1}^λ

S2←{0,1}^λ

S3 = s1 ⊕ s2 ⊕ m_R

Return(s1)

If(U={2})

S1←{0,1}^λ

S2←{0,1}^λ

S3 = s1 ⊕ s2 ⊕ m_R

Return(s2)

If(U={3})

k ←{0,1}^λ

S3 = k ⊕ m_L

Return(s3)

If(U={1,2})

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$S3 = s1 \oplus s2 \oplus m_R$

Return(s1,s2)

If(U={1,3})

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$k \leftarrow \{0,1\}^\lambda$

$S3 = k \oplus m_L$

Return(s1,s3)

If(U={2,3})

$S1 \leftarrow \{0,1\}^\lambda$

$S2 \leftarrow \{0,1\}^\lambda$

$k \leftarrow \{0,1\}^\lambda$

$S3 = k \oplus m_L$

Return(s2,s3)

This would condense back down to the original library and show that both the left and right hand choices are interchangeable.

2) Reconstruct the secret from a 2 out of 10 Shamir secret sharing over Z11, Alice's share was 4,6

And Bobs was (7,3). Also find other 8 shares.

Shamir Secret Sharing algorithm reconstructs the secret using the following formula in which A0 represents the secret

$$f(x) = \sum_{j=0}^{k-1} a_j (x^j \bmod Z) \text{ --------------------- } (*)$$

In this case shmair's will be constructing a line because we only have two points

IE equation of a line $y = mx + c$

Alice's share(4,6) where $x = 4$ and $f(x)$ 6

$6 = A0 + A1(4 \bmod 11)$

Bob's share (7,3)

3 = A0 + A1(7 mod 11)

Have to solve using substitution

A0 = 6 – A1(4)

3 = 6 – 4(A1) + 7(A1)

A1 = -1

A0 = 6-(-1)(4) = 10. Thus the secret code is 10

To calculate the other shares we substitute the values of A0 and A1 with the values of x from 1 to 10 not including 4 and 7 (because we already know those shares)

f(x) = 10 + (-1)x

X = 1 we get share (1,9) and the following shares (2,8), (3,7), (4,6),(5,5),(6,4),(7,3),(8,2),(9,1),(10,0)